

A Neural Approach to ADHD Detection in Children: Enhanced EEG Analysis with Wavelet-Transformer Synergy

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Abstract: The prediction of Attention Deficit Hyperactivity Disorder (ADHD) using Electroencephalogram (EEG) data poses a considerable challenge due to the intricate nature of brain signals and variations among individuals. Early diagnosis is vital for effective management. This study introduces an innovative ADHD prediction system that utilizes advanced EEG data processing techniques along with a transformer-based predictive model to accurately identify ADHD symptoms in children. The research employs an efficient pipeline for EEG data preprocessing and analysis, integrating the wavelet transform to generate informative 2D time–frequency representations, providing a richer description of neural activity compared to raw time-domain data. A transformer encoder is then applied directly to these 2D representations, leveraging the attention mechanism’s strength in modeling global relationships across time and frequency domains. Furthermore, a comparative analysis of various deep learning architectures is conducted to assess their performance in ADHD detection. The findings demonstrate the effectiveness of the proposed ADHD prediction system, showcasing enhanced performance compared to existing approaches by achieving an accuracy and an F1-score exceeding 99%.

1 INTRODUCTION

Attention Deficit Hyperactivity Disorder (ADHD) is one of the most common neurodevelopmental disorders, typically diagnosed during childhood (WHO, 2021). ADHD affects an estimated 5-7% of children globally, leading to challenges in academic performance, social interactions, and emotional regulation (NIMH, 2021). The symptoms are categorized into three main types: inattention, hyperactivity, and impulsivity, although many children may present a combined type, exhibiting traits of both. Neurobiologically, ADHD is associated with dysregulation in the prefrontal cortex and subcortical regions such as the basal ganglia, which play critical roles in attention, behavior control, and executive function. Children with ADHD not only exhibit cognitive and behavioral difficulties but are also more likely to experience comorbidities such as anxiety, depression, and learning disabilities. Early diagnosis and intervention are crucial in managing ADHD to help mitigate its long-term

effects on education, social relationships, and overall quality of life.

Today, clinicians diagnose patients mainly by relying on subjective criteria outlined in the diagnostic and statistical manual of mental disorders, (DSM-5) (APA, 2013). The current methods rely heavily on subjective assessments, including behavior rating scales, interviews with parents and teachers, and clinical observations, all of which may vary across clinicians and patients. However, these evaluations are susceptible to bias and inconsistency, leading to possible delays in the diagnosis and misidentification of this complex disorder. As such, the need for more objective and reliable data-driven analysis-based diagnostic tools has gained prominence, particularly those that incorporate biological markers, such as electroencephalogram (EEG) readings (Cai et al., 2025; Jafari et al., 2024). By detecting ADHD symptoms early, these tools can help healthcare professionals diagnose the disorder at a younger age, leading to earlier interventions and better management of the condition.

Advancements in artificial intelligence, particularly in machine learning (ML) and deep learning (DL), offer significant potential for improving ADHD

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diagnosis by analyzing complex brain patterns. Recent developments in ML have shown promise in automating ADHD diagnosis through various modalities, including handwriting data (Walha et al., 2025a), brain imaging (Begum and Shaik, 2025), electroencephalogram signals (Cai et al., 2025), eye-tracking data (Sharma et al., 2025), and cognitive performance data (Walha et al., 2025b). DL, which excels at identifying intricate patterns in large computer vision datasets (Elbahri and Walha, 2024; Harizi et al., 2025; Mallek et al., 2017), is being explored as an effective tool to detect ADHD more precisely by recognizing unique neural activity signatures in children with the disorder (Moghaddari et al., 2020). In particular, different DL models have demonstrated their ability to analyze EEG signals and detect neurophysiological markers indicative of ADHD (Latifi et al., 2024; Hassan and Singhal, 2024; Chen et al., 2019). Especially, convolutional neural networks (CNNs) excel in capturing local spatial patterns through the captured data. However, these networks struggle with effectively handling long-range dependencies and global context. The Vision Transformer (ViT) has emerged as a highly effective framework for computer vision tasks (Guo et al., 2025; Walha et al., 2024). Our motivation behind using the principle of ViT arises from the need to capture both local and global features in visually transformed EEG data for improved performance in the ADHD prediction task.

The main purpose of this research study is to enhance the diagnosis of ADHD in children by leveraging advanced signal processing and deep learning techniques for EEG data analysis. Especially, the main contributions of this work are threefold: (1) an efficient EEG data preprocessing and analysis pipeline using wavelet transform, (2) the integration of a transformer encoder for complex pattern detection in the EEG data in order to detect ADHD signs, and (3) a comparative study between various deep learning architectures for EEG-based ADHD assessment. More precisely, we introduce a novel ADHD prediction system that relies on robust EEG data processing and analysis, enabling the adoption of wavelet transform and ViT principle for ADHD assessment. In fact, the proposed system integrates a Haar wavelet-based transformation on EEG segments which involve information across a wide range of frequencies and the ability to analyze these frequencies at different resolutions can provide more detailed insights into brain activity. Moreover, the proposed system involves a transformer encoder that is particularly suited for ADHD detection task as it utilizes self-attention mechanisms, allowing it to capture pertinent patterns and intricate dependencies in the EEG

data. This proposed approach, combining wavelet transform and ViT, allows the proposed system to effectively capture from EEG signals pertinent features associated with ADHD, thus improving the classification performance. To the best of our knowledge, this work is the first to explore the effectiveness of the wavelet-based transformation combined with the principle of ViT to predict ADHD from EEG signals.

The rest of this paper is organized as follows: Section 2 outlines related works on ADHD disease prediction. Section 3 details the proposed system, discussing the methodology and the rationale behind our proposition. Section 4 presents the experimental study and discusses results. Section 5 concludes the paper with a summary of the entire work and highlights emerging aspects for future research.

2 RELATED WORK

Numerous studies have explored the use of EEG to identify atypical neural patterns in individuals with ADHD. For instance, Parashar et al. (Parashar et al., 2021) analyzed EEG data from 60 ADHD children and 60 healthy controls, classifying ADHD using three classifiers: AdaBoost, Random Forest, and Support Vector Machine (SVM). The best performance was achieved with the AdaBoost classifier, which obtained an accuracy of 84% with a sensitivity of 96%. A recent study (Maniruzzaman et al., 2023) employed channel and feature selection methods to identify optimal EEG channels for ADHD detection, yielding to 28 features extracted using LASSO logistic regression. The Gaussian process classification achieved the highest performance. Another study (Alim and Imtiaz, 2023) developed a Gaussian SVM-based model with linear EEG features extracted from the first four sub-bands, achieving a mean accuracy of 94%.

Recent advancements in DL and EEG signal analysis offer promising approaches for improving ADHD detection. For instance, Chen et al. (Chen et al., 2019) proposed a CNN-based framework to identify ADHD by analyzing EEG data from 50 ADHD children and 51 controls, achieving an accuracy of 94.67%. (Moghaddari et al., 2020) proposed a CNN-based approach for ADHD detection using EEG signals recorded during continuous mental tasks. EEG data were preprocessed, segmented, transformed into RGB images via frequency band extraction, and classified with a 13-layer CNN, achieving over 90% accuracy. Similarly, LSTM networks were applied in (Chang et al., 2022) to detect ADHD, achieving 90.50% accuracy. In (Jahani and Safaei, 2024), residual network (ResNet) is applied and demonstrated su-

perior accuracy compared to CNN, highlighting the potential of cognitive-inspired models in ADHD diagnosis. Chauhan et al. (Chauhan and Choi, 2024) used mutual information to select relevant EEG features. Ten classifiers were trained, and their outputs were combined using a stacked classifier with a multilayer perceptron (MLP) as the meta-classifier.

Many research efforts in EEG classification and analysis focus primarily on processing raw signal data. Traditional methods exploit the temporal and frequency characteristics of EEG signals to detect patterns associated with conditions such as ADHD (Chauhan and Choi, 2024). However, a growing trend involves converting EEG signals into 2D image representations, enabling the application of DL models like CNNs (Yilmaz et al., 2023; Ch Vidyasagar et al., 2024). This transformation allows researchers to uncover complex patterns and spatial relationships that might not be as evident in the raw signal format.

The reviewed studies demonstrate the effectiveness of various models in achieving high classification accuracy. These advancements indicate significant progress in developing objective and reliable tools for ADHD diagnosis, with potential implications for clinical practice and early intervention. However, different existing works often face challenges when dealing with the high-dimensional and complex nature of EEG signals, where both local details and global information are crucial for accurate interpretation. Recently, there exists a growing body of work that combines or merges DL techniques with traditional data analysis methods (Yao et al., 2022; Harizi et al., 2024; Roozbehi et al., 2025). This hybrid approach leverages the strengths of both paradigms, where DL models excel at capturing complex patterns and representations from large datasets, while classical techniques like wavelets offer powerful tools for analyzing signals at multiple scales, providing both frequency and temporal localization. By integrating these methods, researchers aim to enhance features learning and hence models performance in order to overcome certain limitations inherent in each individual approach. So, further interesting research is essential to enhance the robustness and applicability of existing methods.

3 PROPOSED ADHD PREDICTION SYSTEM

3.1 System Overview

In this section, we present an overview of the proposed ADHD prediction system, which aims to assess whether an individual has ADHD relying on effi-

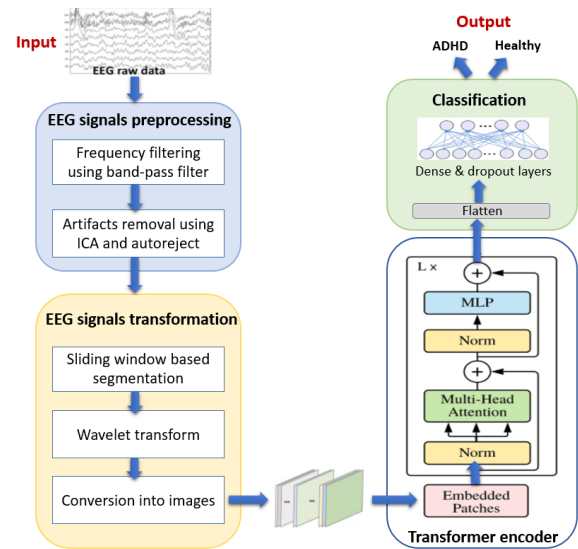


Figure 1: Overview of the proposed ADHD's disease prediction framework.

cient EEG data analysis. The system begins with EEG data acquisition, where signals are collected from participants during specific tasks. After that, the collected signals undergo several preprocessing steps to ensure they are in an optimal state for analysis. These steps include filtering to remove noise, the independent component analysis (ICA) for artifact removal, and the autoreject technique for further cleaning. Following these preprocessing techniques, the EEG data is segmented using a sliding window approach, dividing the signals into shorter, manageable segments for analysis. After segmentation, we introduce a wavelet transform to capture important time-frequency characteristics, which are essential for identifying patterns associated with ADHD. Once the wavelet-based transformation is complete, the EEG data are converted into image representations. Specifically, each window of the EEG data is transformed into an informative 2D time-frequency representation. These wavelet-based maps capture both temporal dynamics and spectral variations, providing a richer description of neural activity compared to raw time-domain EEG signals. This conversion enables the application of image-based DL models. In this study, we leverage the ViT, a state-of-the-art model designed to efficiently process image data. The transformer encoder is particularly suited for this task as it utilizes self-attention mechanisms, allowing it to capture pertinent features and intricate dependencies in the data, which can improve the classification accuracy. This design step introduces a novel and powerful perspective in EEG-based learning, as it leverages the attention mechanism's strength in modeling global rela-

tionships across time and frequency domains. The final step of the system involves classifying the extracted features into two categories: individuals with ADHD and healthy controls. By utilizing this approach, we take advantage of the wavelet transform and ViT's strengths in image classification to achieve high performance in detecting the neurodevelopmental disorder ADHD. An overview of the proposed approach is depicted in Fig. 1. More details about each module of our proposition are given in the following subsections.

3.2 EEG Signals Preprocessing

3.2.1 Frequency Filtering

The EEG preprocessing starts with a crucial phase: filtering. This step is vital for improving and preparing the raw EEG signals, making them appropriate for further analysis. Filtering aims to extract the relevant frequency components while eliminating any extraneous noise and artifacts. First, we apply a notch filter with a frequency of 50 Hz to address power line interference and electrical noise that can contaminate the EEG signal. This filter effectively eliminates the 50 Hz frequency component, which is commonly associated with electrical disturbances from power sources. Following the notch filter, we implement a sixth-order Butterworth band-pass filter with a frequency range of 0.1 to 60 Hz in order to remove low-frequency drift and high-frequency noise, preserving the signal's relevant frequency components. The band-pass filter ensures that only the frequencies crucial for ADHD classification are retained, while unwanted artifacts and irrelevant noise are removed.

By applying these filtering techniques, we enhance the quality of the EEG data, making it well-suited for the detailed and intricate analyses required in our study. This preprocessing step is fundamental in ensuring that the subsequent data processing and classification are based on clean, high-quality EEG signals.

3.2.2 Artifacts Removal

To refine the EEG data, Independent Component Analysis (ICA) was used to decompose the signals and remove artifacts associated with eye movements and muscle activity. ICA was configured with 15 components and a maximum of 800 iterations. Channels linked to eye artifacts, such as 'Fp1' and 'F8', were excluded. Following ICA, the autoReject technique was applied to discard bad epochs, ensuring that only high-quality data segments were retained. The cleaned EEG signals, now free of artifacts, was saved for further analysis.

3.3 EEG Signals Transformation

3.3.1 Data Segmentation

In this transformation phase, the filtered and cleaned EEG data is segmented using a sliding window-based approach. This involves applying a window size of 64 samples with a stride of 32 samples to create overlapping segments from the continuous EEG signals. By overlapping the windows, this step ensures smoother transitions across segments and minimizes the loss of relevant temporal information that could occur with non-overlapping slicing. This segmentation strategy not only increases the number of training instances but also helps capture transient neural patterns and short-duration variations that may be critical for distinguishing cognitive states such as ADHD.

3.3.2 Wavelet Transform

After completing the EEG data segmentation step, and motivated by the theory of wavelet, we propose to integrate a Haar wavelet-based transformation step. In fact, EEG signals often contain transient that consists of short-duration events, such as spikes and sharp waves, which are difficult to capture using traditional methods like the Fourier transform. The proposed wavelet decomposition is particularly suited for this purpose, as wavelets can identify these transient features in the EEG signals, providing a better representation of cognitive events. Moreover, this transformation allows for analyzing the segments of EEG signal in both the time domain and the frequency domain simultaneously. This is particularly useful for EEG data, which exhibit transient events or changes in frequency that can be localized in time.

In this work, the EEG data segments, previously processed through sliding windows, undergo a wavelet decomposition using the Haar wavelet. This transformation breaks down each segment into multiple frequency bands, capturing both low and high-frequency components across different scales. The resulting wavelet coefficients, representing various levels of detail, are organized and saved in a structured format for each data segment. This approach enriches the data with a multi-resolution perspective which means that the EEG segments can be analyzed at different levels of detail or scales, enhancing the subsequent classification and analysis stages by providing detailed time-frequency features. This Haar wavelet-based transformation step is important for predicting ADHD from EEG segments because these latter involve information across a wide range of frequencies and the ability to analyze these frequencies at different resolutions can provide more detailed insights into brain activity.

3.3.3 Conversion of EEG Data into Image Representations

In this step, the wavelet-transformed EEG signals are converted into image representations. This conversion can serve as a powerful step that enables the use of advanced image analysis and DL architectures like ViT. This allows the treatment of the EEG signal as a sequence of image patches, enabling better capture of global contextual information. Especially, this transformation into 2D representations can help reveal hidden patterns in the EEG data, and then facilitate better feature extraction for the prediction of ADHD. In fact, the brain's electrical activities can exhibit temporal as well as spatial patterns that are more evident when viewed as an image, aiding in the detection of specific conditions.

Each processed EEG segment is initially represented as a single-channel grayscale image, capturing the essential time-frequency information from the wavelet decomposition. To prepare these grayscale images for compatibility with ViT and other RGB-based models, each grayscale image is expanded to a three-channel format by duplicating the single-channel data across RGB channels. This transformation enables effective utilization of image-based DL techniques to detect patterns in EEG data associated with ADHD.

3.4 Classification Using Vision Transformer-Based Architecture

For the ADHD prediction task using EEG signals, we propose to adapt the ViT architecture, traditionally used in image classification, to model the complex patterns within EEG data. While ViT has demonstrated exceptional performance in image processing by leveraging its self-attention mechanism to capture global dependencies between patches, we aim to explore its potential for analyzing relevant visual features within the wavelet-transformed EEG images. More precisely, the proposed ViT-based architecture processes EEG-derived images by treating them as a sequence of patches, enabling it to learn complex spatial relationships within the signals. By transforming these inputs into sequences of embeddings, our proposed deep model can effectively learn the intricate patterns and relationships embedded in the EEG data that are critical for accurate ADHD prediction. In particular, the transformer encoder is responsible for processing this sequence of image patch embeddings, leveraging its self-attention mechanism to understand the relationships and context between different patches in the wavelet-transformed EEG image. This encoder consists of multiple layers of self-

Table 1: Configuration of the proposed architecture.

Layer	Configuration
Input image	Size: 224×224
Transformer Encoder Layers	12
Hidden Dimension	768
Number of Attention Heads	12
MLP Head Hidden Layers	2 (256, 64 units)
Dropout	Rate: 0.3
Dense	Units: 256
Dropout	Rate: 0.3
Dense	Units: 64
Dense	Units: 2

attention and feedforward networks. A total of 12 layers are used in this work. Each layer builds upon the representations learned by the previous one, enabling the proposed model to capture increasingly abstract features and complex relationships between the wavelet-transformed EEG image patches.

Unlike CNNs, which rely on local filters, the transformer encoder can capture global patterns across the entire image, making it well-suited for identifying subtle differences between ADHD and healthy control samples. Specifically, the self-attention mechanism involved in the transformer encoder enables the model to understand the contextual dependencies between patches, overcoming the limitations of local receptive fields in CNN-based architectures.

The configuration of the proposed ViT-based deep architecture, detailing each layer and its specific parameters, is presented in Table 1.

The proposed ViT-based architecture provides a robust and efficient solution to the ADHD detection problem, offering strong generalization capabilities by taking advantage of both the wavelet-transformed data and the power of transformer architecture.

4 EXPERIMENTAL STUDY

This section presents the experimental evaluation conducted to assess the effectiveness and robustness of the proposed ADHD prediction system.

4.1 Experimental Framework

In this study, a publicly available EEG dataset (Nasrabadi et al., 2020) from IEEE DataPort is used, consisting of 60 healthy controls and 61 children with ADHD, aged from 7 to 12 years. Diagnoses were made by a psychiatrist following DSM-IV criteria. Healthy controls had no history of neurological or psychiatric disorders. EEG recordings were taken during a visual attention task, where children counted

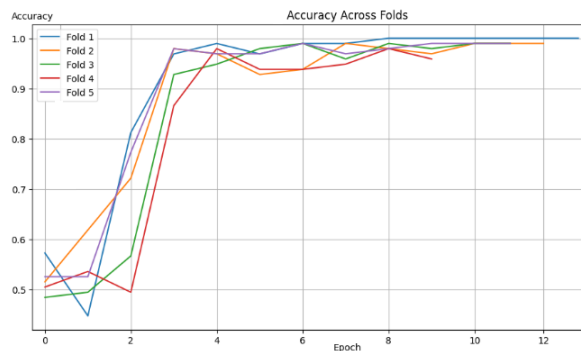


Figure 2: Evolution of the accuracy across folds during the training process of the proposed ADHD prediction model

cartoon characters in images containing 5–16 characters. The task was continuous, and EEG recording duration depended on the child’s response speed.

Accuracy, precision, recall, and F1-score are the metrics which are used in our experimental study.

4.2 Performance Evaluation of the Proposed ADHD Prediction System

In our experimental study, we applied the k-fold cross-validation technique to partition the used dataset. In this approach, the data is divided into k distinct subsets (or folds), where k-1 folds are used for training the model, and the remaining fold is reserved for testing. The overall performance of the model is determined by averaging the results across all k test folds. In our case, we utilized 5-fold cross-validation, meaning the model was trained five times on 80% of the data and tested on the remaining 20% each time. Fig. 2 presents the results for each fold, by using a learning rate of 0.0001, a batch size of 32, Adam optimizer, and 200 epochs, along with the application of early stopping mechanism. As illustrated in this figure, the accuracy plots during training for each fold reflect our model’s performance dynamics over time.

4.3 Exploring the Efficiency of Different Deep Models for ADHD Prediction

To evaluate the effectiveness of the proposed transformer-based ADHD predictor model, we compare its performance with different deep learning models, including CNN (LeCun et al., 2015), the Graph Neural Network (GNN) (Fey and Lenssen, 2019), and the Swin Transformer (ST) (Liu et al., 2021). In order to ensure a fair comparison, we utilized the same preprocessed EEG data. The training hyperparameters, such as learning rate and batch size, were kept consistent with the settings used for the pro-

posed model.

As presented in Table 2, results indicate that the proposed transformer-based model outperformed the other models involved in this study, achieving a higher accuracy of 99.20% compared to the CNN’s 98.37%. Following this, the GNN and Swin Transformer models also demonstrated strong performance, albeit with slightly lower accuracy. This suggests that the ability of the proposed transformer encoder to capture spatial dependencies through self-attention mechanisms is particularly advantageous for EEG classification tasks, especially the EEG-driven ADHD prediction task.

4.4 Comparison with State-of-the-Art Methods

In this section, we present a thorough analysis of the performance of the proposed system for ADHD disorder prediction, in comparison to other state-of-the-art systems. The comparison study, as shown in Table 3, offers insightful information about the accuracy achieved by each system, allowing us to make defensible judgments about the efficiency of our proposed ADHD prediction system. Furthermore, we also highlight studies that have utilized the same EEG dataset (Nasrabadi et al., 2020) as our work, providing a more direct comparison of performance and demonstrating the effectiveness of our approach in this specific context.

According to Table 3, the comparative analysis reveals significant variations in the effectiveness of different ADHD prediction methodologies. For instance, the proposed system, which relies on wavelet-based EEG data analysis with transformer encoder-based model, outperforms the other systems involved in this experiment, achieving an impressive accuracy of 99.20% and demonstrating the superior capability of advanced transformer architecture in handling complex EEG data. In contrast, existing systems, such as those based on Gaussian SVM and Bernoulli Naive Bayes, while effective, attain lower accuracy rates of around 93% and 96%, respectively. Notably, deep learning models such as CNNs and ResNet, as well as the innovative EPSPatNet86, show robust performance with accuracies exceeding 90%, indicating their substantial role in EEG-driven ADHD detection. These results underscore the promise of integrating the proposed advanced transformer-based predictor model and the proposed sophisticated EEG data analysis techniques in improving ADHD diagnostic accuracy. The findings highlight the impact of model choice and data handling approach in achieving high performance.

Table 2: Comparative study between different deep models for ADHD prediction.

Deep model	Accuracy	Precision	Recall	F1-score
GNN (Fey and Lenssen, 2019)	97.50%	94.70%	97.50%	96.07%
CNN (LeCun et al., 2015)	98.37%	98.49%	98.37%	98.42%
ST (Liu et al., 2021)	92.56%	92.57%	92.56%	92.56%
Proposed model	99.20%	99.27%	99.20%	99.23%

Table 3: Comparative study between different ADHD prediction systems evaluated on the same EEG dataset.

Study	Used model	Accuracy
(Moghaddari et al., 2020)	CNN-based model	98.48%
(Parashar et al., 2021)	AdaBoost, RF, SVM classifiers	84.00%
(Tanko et al., 2022)	EPSPatNet86 pattern learning network	97.19%
(Esas and Latifoğlu, 2023)	CNN-based model	87.00%
(Maniruzzaman et al., 2023)	Gaussian process-based classifier	97.53%
(Alim and Imtiaz, 2023)	Gaussian SVM classifier	93.20%
(Ahire et al., 2023)	Bernoulli Naive Bayes classifier	96.00%
(Jahani and Safaei, 2024)	ResNet-based model	96.80%
(Deshmukh et al., 2024)	Random Forest classifier	82.81%
(Chauhan and Choi, 2024)	Stacked classifier	92.00%
(Cai et al., 2025)	Topological data analysis-based model	78.27%
(Ullah et al., 2025)	Gated Recurrent Unit-based model	93.20%
(Sanchis et al., 2025)	MCRNet model	94.87%
Proposed system	Wavelet and transformer-based model	99.20%

5 CONCLUSION AND OPEN ISSUES

In this study, we explored the adaptation of advanced deep learning architecture like ViT for ADHD prediction using EEG signals. By implementing a comprehensive wavelet-based data processing pipeline, we successfully transformed raw EEG data into a format suitable for classification. The proposed system achieved interesting results, demonstrating its effectiveness in distinguishing between ADHD and non-ADHD subjects based on EEG patterns. These results underscore the potential of leveraging sophisticated signal analysis techniques and advanced deep learning architectures for a precise and an early detection of children with ADHD, paving the way for more reliable assessment methods in clinical settings.

The exceptional performance of the proposed system provides valuable insights for future research. In particular, future work could explore the integration of multi-modal data (e.g., behavioral assessments and genetic information) to enhance predictive performance. Additionally, real-time implementation of the system in clinical environments presents challenges that require exploration, including optimization for computational efficiency and user-friendliness. Addressing these open issues will be interesting for advancing the practical application of EEG-based ADHD diagnosis in children.

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